

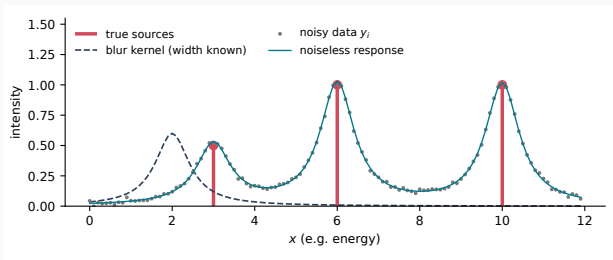
Bayesian spectral deconvolution with Hamiltonian Monte Carlo

How many peaks are in my spectrum — and where?

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1. The problem: blurred, noisy spectra



Measured spectrum = sum of K blurred sources + noise:

$$y_i = \sum_{k=1}^K I_k \phi(x_i - \mu_k) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

with a Lorentzian kernel of known width γ :

$$\phi(x - \mu) = \frac{1}{1 + ((x - \mu)/\gamma)^2}$$

Unknowns: $\theta = (I_1..I_K, \mu_1..\mu_K)$

Two questions: what are θ , and what is K ?

2. Bayes: a distribution over parameters, not a single fit

$$p(\boldsymbol{\theta} | y, K) = \frac{p(y | \boldsymbol{\theta}, K) p(\boldsymbol{\theta} | K)}{Z_K}, \quad Z_K = \int \mathcal{L}(\boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

posterior = likelihood $\times$ prior / evidence

Likelihood (Gaussian noise)

$$\log \mathcal{L} = -\frac{1}{2\sigma^2} \sum_i \left(y_i - \sum_k I_k \phi(x_i - \mu_k) \right)^2 + \text{const}$$

Priors (uniform boxes)

$$I_k \sim U(0.2, 1.2), \quad \mu_k \sim U(1, 11)$$

Why sample?

- Posterior of $\boldsymbol{\theta}$ is non-Gaussian and multimodal
- We want error bars, not just a best fit
- Z_K needs an integral over $2K$ dimensions

$\Rightarrow$ Markov chain Monte Carlo: draw samples $\boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta} | y, K)$

3. Hamiltonian Monte Carlo: let the posterior push the sampler

Treat θ as a position in potential U , add a fictitious momentum $\mathbf{p}$:

$$U(\theta) = -\log \mathcal{L} - \log p(\theta), \quad H = U(\theta) + \frac{1}{2} \mathbf{p}^\top \mathbf{p}$$

Leapfrog steps (step size ϵ):

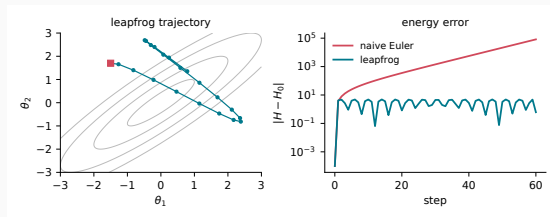
$$\mathbf{p} \leftarrow \mathbf{p} - \frac{\epsilon}{2} \nabla U,$$

$$\theta \leftarrow \theta + \epsilon \mathbf{p},$$

$$\mathbf{p} \leftarrow \mathbf{p} - \frac{\epsilon}{2} \nabla U$$

Accept with $\min(1, e^{-\Delta H})$.

Long, informed moves instead of random-walk steps; the gradient ∇U points where the fit improves.



Toy 2D correlated Gaussian. Leapfrog nearly conserves H , so proposals far away are still accepted (notebook: $\approx 80\%$ at $\beta = 1$).

4. Parallel tempering: escaping local optima

Run chains at inverse temperatures

$$0 \leq \beta_1 < \dots < \beta_L = 1:$$

$$p_\beta(\theta | y) \propto \mathcal{L}(\theta)^\beta p(\theta)$$

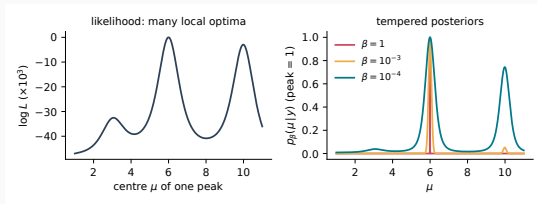
- $\beta = 0$: prior (explores everything)
- $\beta = 1$: true posterior (the answer)

Neighbouring chains swap states with probability

$$\min\left\{1, e^{(\beta_i - \beta_j)(E_i - E_j)}\right\}, \quad E = -\log \mathcal{L}$$

Hot chains find new modes; swaps hand them to the cold chain.

Notebook: 36 temperatures, HMC in each chain, swap every 10 steps, mean swap acceptance ≈ 0.6 – 0.7 .



Profile of the likelihood over one peak position for our data: several optima at $\beta = 1$, a nearly flat landscape at small β .

5. Choosing K : the evidence from the temperature ladder

Tempering also gives Z_K for free. With

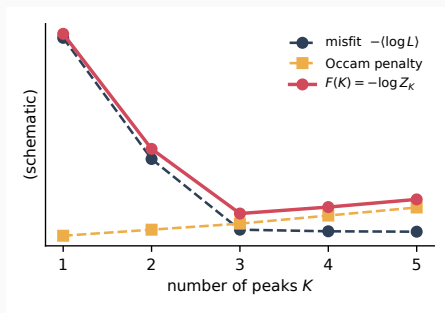
$$Z(\beta) = \int \mathcal{L}^\beta p d\theta, \quad Z(0) = 1:$$

$$\log Z_K = \sum_{l=1}^{L-1} \log \frac{Z(\beta_{l+1})}{Z(\beta_l)}, \quad \frac{Z(\beta_{l+1})}{Z(\beta_l)} = \left\langle \mathcal{L}^{\beta_{l+1}-\beta_l} \right\rangle_{\beta_l}$$

each term is a plain average over samples of chain l .

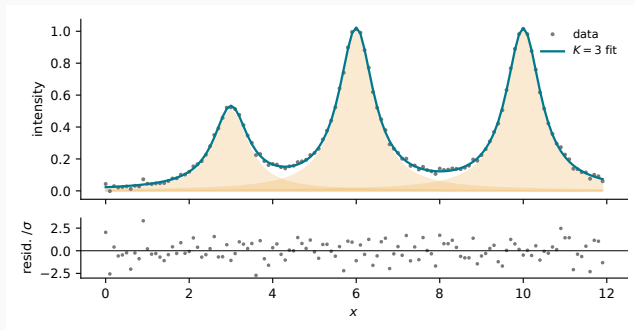
Stochastic complexity $F(K) = -\log Z_K$; choose $K^* = \arg \min_K F(K)$.

Automatic Occam's razor: extra peaks fit slightly better but spread the prior over more volume, so Z_K drops.



Schematic of the trade-off. Least-squares χ^2 on our data: $K=2$: 19 700, $K=3$: 138 (120 points), $K=4$: 137.

6. Result on the synthetic spectrum



Three true peaks at $\mu = 3, 6, 10$;
evidence compared for $K = 2, 3, 4$.
 $K^* = 3$ selected. Posterior (cold chain):

μ_k	$\pm$	l_k
3.004	0.003	0.49
6.001	0.002	1.00
10.001	0.002	0.99

Take-aways

- HMC: efficient moves in $2K$ dims
- Tempering: no local traps, gives Z_K
- K chosen without heuristics